Supplementary material

Strong at home, going global: China's rise in immune checkpoint therapy patents, 2010–2023

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All tables are also available as CSV files at github.com/descartesmbogning/immune-checkpoint-patent-explorer (folder paper/tables).

Table S1. Lens.org search query

Search run on Lens.org on 28 January 2024 (patent documents, all jurisdictions). Spelling variants of some names are shortened here.

```
(Abatacept OR atezolizumab OR Ipilimumab OR lisavanbulin OR Nivolumab OR
pembrolizumab OR "PHI-101"
 OR quemliclustat OR relatlimab OR sotorasib OR spartalizumab OR
Camrelizumab OR Tislelizumab OR avelumab
 OR durvalumab OR cemiplimab OR Tremelimumab OR Toripalimab OR
Dostarlimab OR Retifanlimab OR Lambrolizumab)
OR (Sintilimab OR Vopratelimab OR Spartalizumab OR Camrelizumab OR
Tislelizumab OR "INCMGA00012"
 OR Acrixolimab OR Cosibelimab)
OR ("Checkpoint Inhibitors" OR "Immune Checkpoint Inhibitor" OR "Immune
Checkpoint Blockers"
 OR "Checkpoint Blockers" OR "Immune Checkpoint Blockade" OR "Immune
Checkpoint Inhibition" OR "Checkpoint Inhibition")
OR ("PD-L1 Inhibitor(s)" OR "Programmed Death-Ligand 1 Inhibitors" OR
"PD-1-PD-L1 Blockade"
 OR "CTLA-4 Inhibitor(s)" OR "Cytotoxic T-Lymphocyte-Associated Protein
4 Inhibitor(s)" OR "PD-1 Inhibitor(s)"
 OR "Programmed Cell Death Protein 1 Inhibitor(s)" OR "anti-PD-1" OR
"anti-PD-L1" OR "anti-CTLA-4")
OR (yervoy OR keytruda OR opdivo OR Imjudo OR Tecentriq OR Bavencio OR
Imfinzi OR Libtayo OR Jemperli
 OR Zynyz OR Loqtorzi)
```

Table S2. Keyword rules for checkpoint targets and therapeutic strategies

Rules are case-insensitive regular expressions applied to the title and abstract. A document can match several rules. Documents with no target match are "Not stated"; documents with no strategy match are "Single agent / not stated".

- **PD-1** (target): `\bpd-?1\b|programmed (?:cell )?death[- ]1\b|programmed death receptor 1|\bcd279\b`
- **PD-L1** (target): `\bpd-?l1\b|programmed (?:cell )?death[- ]ligand[- ]1|\bb7-?h1\b|\bcd274\b`
- **CTLA-4** (target): `\bctla-?4\b|cytotoxic t[- ]lymphocyte[- ]associated|\bcd152\b`
- **LAG-3** (target): `\blag-?3\b|lymphocyte[- ]activation gene[- ]3|\bcd223\b`
- **TIM-3** (target): `\btim-?3\b|\bhavcr2\b`
- **TIGIT** (target): `\btigit\b`
- **Other checkpoint** (target): `\bvista\b|\bb7-?h3\b|\bcd47\b|\bnkg2a\b|\bido1?\b|\bcd73\b|\ba2ar\b|\bsiglec-?15\b|\bbtla\b`
- **Combination therapy** (strategy): `\bcombination|\bcombined (?:with|for|use|administration|application|treatment)|\bin combination|co-?administ|joint(?:ly)? (?:use|administ)|synergi|in conjunction with|used together`
- **Bispecific / multispecific** (strategy): `bi-?specific|multi-?specific|tri-?specific|bifunctional`
- **Formulation / delivery** (strategy): `formulation|subcutaneous|dosage form|dosing regimen|lyophili|sustained release|nanoparticle`
- **Biomarker / diagnostic** (strategy): `biomarker|predict(?:ing|ive|s)? (?:the )?response|companion diagnostic|\bdiagnos`
- **Cell therapy** (strategy): `\bcar-?t\b|chimeric antigen receptor|\bcar-?nk\b|cell therapy|adoptive|\btil\b`
- **Vaccine / oncolytic virus** (strategy): `vaccin|oncolytic`
- **Antibody–drug conjugate** (strategy): `antibody[- ]drug conjugate|\badc\b`
- **Small molecule** (strategy): `small molecule|compound of formula|compounds of formula|\bformula \(i\)`

Table S3. Agreement between keyword labels and hand coding

Random sample of 200 documents from 2010–2021 (100 with a Chinese applicant, 100 with a US applicant), coded by the author from titles and abstracts without seeing the rule labels. Precision: share of rule “yes” labels that the coder also marked yes. Recall: share of coder “yes” labels that the rule found. Kappa: Cohen’s kappa. Ten of 1,200 decisions differed.

Label	Group	Coded yes	Rule yes	Agreement (%)	Precision (%)	Recall (%)	Kappa
PD-1	All (n = 200)	101	100	99.5	100.0	99.0	0.99
PD-1	China (n = 100)	56	56	100.0	100.0	100.0	1.00
PD-1	US (n = 100)	45	44	99.0	100.0	97.8	0.98
PD-L1	All (n = 200)	57	58	99.5	98.3	100.0	0.99
PD-L1	China (n = 100)	35	35	100.0	100.0	100.0	1.00
PD-L1	US (n = 100)	22	23	99.0	95.7	100.0	0.97
CTLA-4	All (n = 200)	28	28	100.0	100.0	100.0	1.00
CTLA-4	China (n = 100)	14	14	100.0	100.0	100.0	1.00
CTLA-4	US (n = 100)	14	14	100.0	100.0	100.0	1.00
LAG-3 / TIM-3 / TIGIT	All (n = 200)	6	7	99.5	85.7	100.0	0.92
LAG-3 / TIM-3 / TIGIT	China (n = 100)	2	2	100.0	100.0	100.0	1.00
LAG-3 / TIM-3 / TIGIT	US (n = 100)	4	5	99.0	80.0	100.0	0.88
Combination therapy	All (n = 200)	93	93	97.0	96.8	96.8	0.94
Combination therapy	China (n = 100)	35	34	95.0	94.1	91.4	0.89
Combination therapy	US (n = 100)	58	59	99.0	98.3	100.0	0.98
Bispecific / multispecific	All (n = 200)	17	18	99.5	94.4	100.0	0.97
Bispecific / multispecific	China (n = 100)	11	11	100.0	100.0	100.0	1.00
Bispecific / multispecific	US (n = 100)	6	7	99.0	85.7	100.0	0.92

Table S4. Targets and strategies by applicant country, 2010–2021

Documents with Chinese (and no US) applicants (n = 780) and US (and no Chinese) applicants (n = 962). Shares (%) with Wilson 95% confidence intervals; p from a two-sided two-proportion z-test. These values are plotted in Figure 3.

Label	China, % (95% CI)	United States, % (95% CI)	Difference (points)	p
PD-1	55.5 (52.0–59.0)	42.9 (39.8–46.1)	+12.6	<0.001
PD-L1	44.6 (41.2–48.1)	30.5 (27.6–33.4)	+14.2	<0.001
CTLA-4	8.7 (6.9–10.9)	14.6 (12.5–16.9)	-5.8	<0.001
LAG-3, TIM-3 or TIGIT	2.6 (1.7–3.9)	4.7 (3.5–6.2)	-2.1	0.021
Combination therapy	33.5 (30.2–36.8)	53.8 (50.7–57.0)	-20.4	<0.001
Bispecific / multispecific	10.6 (8.7–13.0)	4.2 (3.1–5.6)	+6.5	<0.001
Biomarker / diagnostic	10.0 (8.1–12.3)	7.9 (6.4–9.8)	+2.1	0.125
Cell therapy	2.8 (1.9–4.2)	4.0 (2.9–5.4)	-1.1	0.199
Formulation / delivery	3.6 (2.5–5.1)	6.2 (4.9–7.9)	-2.6	0.012
Small molecule	6.5 (5.0–8.5)	2.7 (1.9–3.9)	+3.8	<0.001

Table S5. PD-1 or PD-L1 × VEGF bispecific documents

Documents with a PD-1 or PD-L1 label, a bispecific or multispecific label and the term “VEGF” in the title or abstract (all application years). Office: WO = international PCT application.

Year	Office	Applicant country	Title	Lens ID
2019	CN	China	Anti-PD-L1/VEGF bifunctional antibody and application thereof	006-156-871-991-364
2019	CN	China	Anti-VEGF-anti-PD1 bispecific antibody with brand-new sequence	016-976-152-765-295
2019	WO	China	Anti-vegf-anti-pd1 bispecific antibody with new-type structure	034-989-868-512-248
2019	EP	China	Anti-pd-1/anti-vegf natural antibody structure-like heterodimeric form bispecific antibody and preparation thereof	076-646-730-845-444
2019	CN	China	Anti-VEGF and anti-PD1 bispecific antibody with novel structure	169-312-582-824-538
2020	CN	Other	Anti-PD-L1/VEGF/TGF-beta multispecific antibody and application thereof	025-060-456-173-340
2020	EP	China	Anti-pd-1-anti-vegfa bispecific antibody, pharmaceutical composition and use thereof	149-024-907-093-330
2021	CN	China	Anti-VEGF-anti-PD-L1 bispecific antibody as well as pharmaceutical composition and application thereof	013-938-200-146-717
2021	CN	China	Bifunctional protein aiming at VEGF and TGF-beta and application of bifunctional protein	051-587-902-928-840
2021	CN	China	Anti-PD-1/VEGF bispecific antibody liquid preparation	063-425-249-565-453
2021	WO	China	Bifunctional fusion protein having anticancer activity	136-629-167-747-860
2021	CN	China	Bispecific anti-PD-L1/VEGF antibody and application thereof	150-018-177-029-517
2021	PH	China	Anti-pd-1/vegfa bifunctional antibody, pharmaceutical composition thereof and use thereof	194-462-485-651-006
2022	WO	China	A bispecific anti-pd-l1/vegf antibody and uses thereof	020-528-710-380-396
2022	TW	United States	Methods and compositions for treating cancer	062-033-375-765-385
2022	EP	China	Pharmaceutical combination containing anti-pd-1-anti-vegfa bispecific antibody, and use thereof	135-690-562-217-137
2022	CN	China	Use of antibodies in anti-tumor therapy	189-598-565-664-95X
2023	WO	China	Anti-pd-l1, vegf and egfr trispecific antibody and use thereof	028-805-830-990-985
2023	WO	China	Anti-pd-l1 and vegf bispecific antibody and use thereof	064-378-054-048-904
2023	WO	China	Pharmaceutical composition comprising anti-tigit antibody and anti-pd-1-anti-vegfa bispecific antibody, and use	198-878-549-199-667
2023	WO	China	Pharmaceutical composition and use thereof	198-930-667-352-795